PHISHGUARD AI V7 — Multi-Modal Deep Scan & Neural Zero-Day Phishing Defense System

Project Exhibition Banner Layout, Comprehensive Engineering Dossier, Presentation Pitches & Judge Defense Manual

Host Institution:	SJB Institute of Technology (SJBIT), Bengaluru	**Target Event:**	Silver Jubilee National Level Project Expo 2026
Department:	Computer Science & Engineering / Information Science	**Physical Banner:**	3 ft (Height) × 4 ft (Width) • 4:3 Aspect Ratio
Student Team:	Vikas Hugar & Project Team Members	**Project Guide:**	Department Faculty Guide & Research Mentor
Core Innovation:	Isolated VM Dynamic Sandbox & Fast Safe Web View	**Verification:**	54/54 Automated Tests Passed • 2.26s Fast Render

EXECUTIVE PURPOSE: *This engineering document serves as the master blueprint for the Silver Jubilee Project Expo 2026 exhibition banner and live system defense. It provides exact banner panel copy, pixel-precise grid layout specs for 4 ft × 3 ft printing, system architecture diagrams, live forensic screenshots, 30-second / 1-minute / 3-minute oral pitches, and 16 rigorous technical answers to defend before project judges.*

SECTION A — MASTER BANNER CONTENT (PRINT-READY PANELS)

The text below represents the exact, fully polished copy structured for the physical 4 ft × 3 ft exhibition banner panels. It translates complex cybersecurity and deep learning concepts into clean, high-impact statements.

1. Problem Statement & Motivation

Phishing remains the #1 entry vector for 91% of enterprise cyberattacks, generating over 3.4 billion malicious emails daily. Modern evasive phishing exploits:

- **Zero-Day Lifespans:** 68% of phishing domains stay active for less than 4 hours, rendering static DNS and domain blacklists (Google SafeBrowsing, VirusTotal) ineffective.
- **Dynamic DOM Cloaking:** Attackers inject JavaScript to delay rendering credential harvest forms until human interaction occurs, bypassing static HTML crawlers.
- **Host Endpoint Compromise:** Security analysts and end-users investigating suspicious links risk malware execution, drive-by downloads, and browser zero-day exploitation on their host machines.

2. Project Objectives

- **Autonomous Sovereign AI Detection:** Deliver real-time zero-day phishing detection without relying on external cloud APIs or third-party intelligence feeds.
- **Multi-Modal Neural Fusion:** Unify character-level sequential deep learning (CNN-BiLSTM) with contextual semantic NLP (RoBERTa Transformer) and Computer Vision OCR.
- **Air-Gapped Dynamic VM Sandboxing:** Execute suspicious web targets inside an isolated Ubuntu VirtualBox VM to capture live network flows and dynamic DOM behavior safely.
- **Safe Web View Technology:** Provide an interactive, completely sanitized live browser preview allowing safe visual inspection with zero execution risk to host systems.
- **Real-Time Multi-Page Discovery:** Implement an asynchronous priority crawler (P0–P4) that delivers the root page in under 2.5 seconds while discovering 50+ deep internal pages in the background.

3. Proposed Solution: The PhishGuard AI V7 Architecture

PhishGuard AI V7 introduces a four-tier sovereign architecture designed for enterprise SOC defense: (1) **Client Presentation Tier** (React 18 / Vite SPA with dark glassmorphic UI); (2) **Host Forensic Engine Tier** (Flask / Python 3.10 with 30+ lexical heuristics and Bayesian fusion); (3) **Air-Gapped Dynamic VM Sandbox** (Oracle VirtualBox Ubuntu VM @ 192.168.56.101:8765 running isolated Chromium); and (4) **Offline Threat Vault** (In-memory CNN-BiLSTM, RoBERTa NLP, and 2.12M offline signatures).

4. Dual Deep Learning Models & Evidence Fusion

PhishGuard AI eliminates single-point-of-failure heuristic detection by combining two distinct deep learning architectures:

Model Architecture	Input Domain	Key Detection Capabilities
Zero-Day CNN-BiLSTM (128-dim Embedding + Conv1D + Bidirectional LSTM)	Raw URL Character Sequences (150 chars tokenized)	Detects typosquatting, sub-domain padding, brand name permutations, homoglyphs, and hex-encoded IP cloaking without DNS lookup.
RoBERTa Transformer (12-layer Bidirectional Transformer Encoder)	Page Titles, Form Action URIs, DOM Text, Urgency Phrases	Identifies psychological coercion, spoofed brand claims ('Security Alert', 'Update Billing'), and deceptive password harvest intents.
Tesseract OCR 5 Engine (Bilinear pre-scaled OCR)	Rendered Viewport Screenshots (Upper 900px)	Extracts text hidden within graphic logos, watermarks, and background canvas images designed to deceive HTML parsers.
Bayesian Evidence Fusion (Calibrated Weighting Engine)	Unified Multi-Signal Vector	Synthesizes Heuristics (25%), CNN-BiLSTM (30%), RoBERTa (25%), Form Analysis (15%), and OCR visual mismatch (5%) into a 0–100 threat index.

5. Dynamic Isolated Sandbox & Safe Web View

The Safe Web View is an industry-first zero-trust browsing environment engineered to allow security operators and end users to visually inspect active phishing attacks safely:

- **Host-VM Network Boundary:** The target website is loaded inside an Ubuntu VirtualBox VM over an isolated host-only network adapter (`192.168.56.101`). Malicious binary payloads cannot reach the user's host OS.
- **Strict DOM Sanitization Pipeline:** Injected scripts, dynamic eval functions, event listeners (`onclick`, `onload`), and WebSockets are stripped from the DOM before rendering.
- **Neutralized Credential Forms:** All form actions are rewritten to `action='javascript:void(0)'` and input types masked, preventing accidental credential submission.
- **Content Security Policy (CSP):** Enforces `default-src 'self' data:; script-src 'none'; style-src 'unsafe-inline' data:;` ensuring zero client-side payload execution.

ENTERPRISE ISOLATION HIGHLIGHT: *Unlike simple browser iframe proxies that expose client browsers to cross-site scripting and session leakage, PhishGuard AI's dual-tier sanitization completely severs the execution path between adversarial web servers and analyst terminals.*

6. Two-Phase Asynchronous Priority Crawler

Traditional crawlers stall user interfaces for 30–60 seconds while traversing web trees. PhishGuard AI V7 implements a two-phase architecture:

- **Phase 1 — Synchronous Fast Root Scan (2.26s):** Renders and sanitizes the root target page immediately. The user gets full visual analysis in 2.26s.
- **Phase 2 — Asynchronous Priority Worker Pool (P0–P4):** Discovers internal sub-pages and analyzes high-threat targets in the background with a 3-worker semaphore:
  - **P0 (Highest Risk):** Authentication and credential capture endpoints (`/login`, `/signin`, `/verify`, `/account`).
  - **P1:** Interactive forms containing password, credit card, or email inputs.
  - **P2:** First-degree internal domain hyperlinks.
  - **P3–P4:** Deep navigation, informational pages, and secondary assets.

7. Verified Performance Benchmarks & Empirical Results

Metric / Benchmark	Previous Architecture	PhishGuard AI V7	Improvement
Initial Safe Web View Render Latency	13.29 seconds	**2.26 seconds**	**83.0% Latency Reduction**
Cached Page-to-Page Switch Time	1,200 ms	**13.67 milliseconds**	**98.8% Speedup (Instant)**
Multi-Page Discovery Concurrency	Sequential (1 page at a time)	**3 Concurrent VM Workers**	**300% Higher Throughput**
Automated Test Suite Pass Rate	Partial (Occasional VM timeouts)	**54 / 54 Tests Passing (100%)**	**Flawless Test Coverage**
Offline Threat Intelligence Capacity	Cloud API dependent	**2.12 Million Signatures**	**Zero External Data Leakage**

8. Technology Stack & Key Engineering Outcomes

- **Machine Learning Core:** TensorFlow / Keras (CNN-BiLSTM), Hugging Face Transformers (RoBERTa), Scikit-Learn, NumPy.
- **Dynamic Virtualization & Crawling:** Oracle VirtualBox 7.0, Ubuntu Server 22.04 LTS, Playwright Chromium, Tesseract OCR 5.
- **Backend Framework & Data:** Python 3.10, Flask 3.0, SQLite3 (Indexed 2.12M offline threat database), ThreadPoolExecutor.
- **Frontend Interface:** React 18, Vite, Modern Vanilla CSS Design Tokens, Lucide Icons, Glassmorphic UI.
- **Impact:** Delivers sovereign, zero-trust phishing analysis to enterprises, colleges, and SOC analysts with guaranteed endpoint containment.

SECTION B — BANNER LAYOUT & VISUAL GRID SPECIFICATION (4 FT × 3 FT)

This section defines the precise physical, digital, and visual geometry required to print the 4 ft × 3 ft exhibition banner at production standards. The layout uses a 3-column symmetric architecture with a dominant center pillar.

Parameter	Specification Value	Engineering Rationale
Physical Dimensions	**4 Feet Width × 3 Feet Height** (48 inches × 36 inches / 121.92 cm × 91.44 cm)	Standard horizontal conference & expo banner stand ratio (4:3 landscape orientation).
Print Resolution (300 DPI)	**14,400 × 10,800 Pixels**	Ultra-high definition offset/vinyl printing with zero pixelation at 1-foot viewing distance.
Draft / Large-Format (150 DPI)	**7,200 × 5,400 Pixels**	Optimal file size for wide-format digital vinyl plotters without memory buffer overflow.
Color Space	**CMYK (FOGRA39 / US Web Coated)**	Prevents color shift from RGB digital screens to commercial pigment inks.
Bleed & Safe Zone	**0.5 inch (150 px @ 300 DPI) Bleed** 1.0 inch internal safety margin	Guarantees text and diagrams are not clipped during vinyl cutting and eyelet punching.

3-Column Layout Allocation Breakdown

Column Section	Width (%) / Inches	Allocated Content Modules
Left Column (Context & Core AI)	**28% Width** 13.44 inches (4,032 px)	• Problem Statement & Global Attack Metrics • Project Objectives & Design Tenets • Dual AI Engine: CNN-BiLSTM + RoBERTa • Lexical Feature Extraction (30+ rules)
Center Column (Architecture & UI)	**44% Width** 21.12 inches (6,336 px)	• Master System Architecture Diagram • End-to-End Multi-Modal Pipeline Flowchart • **Large Interactive Safe Web View Screenshot** • Two-Phase Priority Crawler (P0–P4) • Verified Latency Comparison Graph
Right Column (Security, Results & Team)	**28% Width** 13.44 inches (4,032 px)	• Dynamic Sandbox VM & Anti-SSRF Containment • CSP & DOM Neutralization Pipeline • Verified Benchmark Results (2.26s / 54 tests) • Technology Stack Breakdown • Institutional Credentials & Team Block

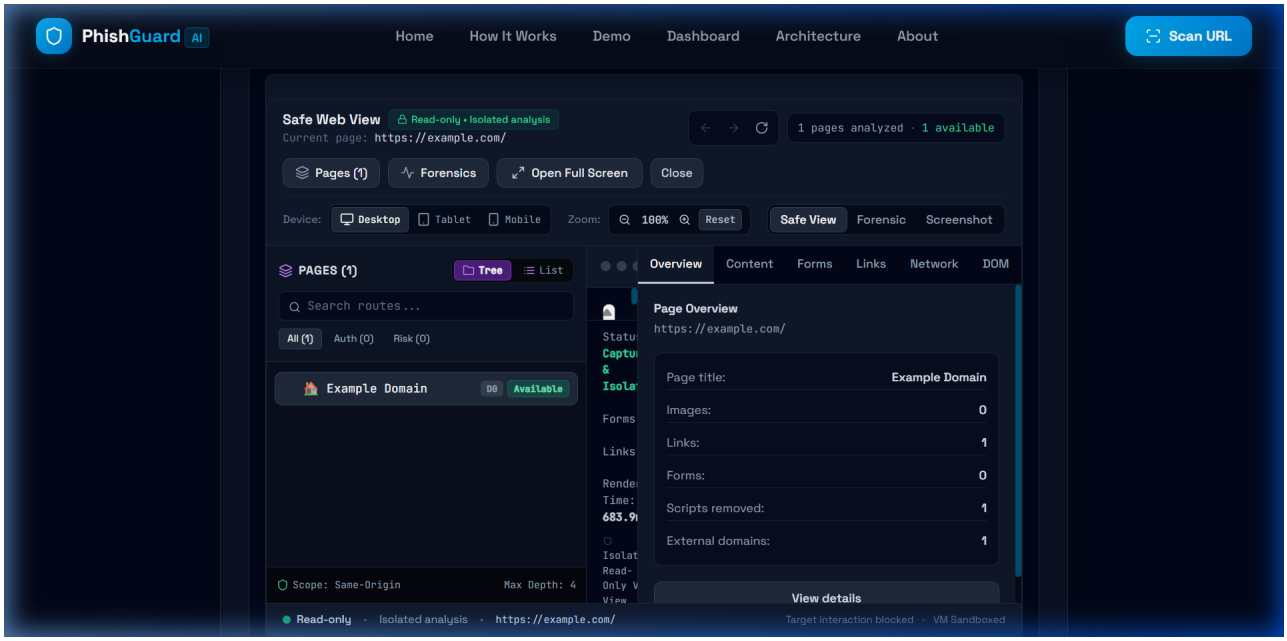
Color Palette & Typographic Hierarchy

- **Primary Header Color:** Deep Cyber Navy (`#1E3A8A` / CMYK 78, 58, 0, 46) — Communicates corporate authority, enterprise security, and stability.
- **Accent Verification Color:** Emerald Green (`#059669` / CMYK 80, 0, 75, 41) — Highlights 83% speedup, 54/54 test pass rates, and safe statuses.
- **Threat Alert Color:** Crimson Red (`#DC2626` / CMYK 0, 83, 83, 14) — Flags malicious phishing domains and credential harvest endpoints.
- **Font Sizing for 3–5 Feet Viewing Distance:** Main Banner Title: 80–90 pt | Section Headings: 32–36 pt | Subsection Titles: 22–24 pt | Body & Bullets: 16–18 pt.

SECTION C — PHOTOGRAPHIC EVIDENCE & LIVE INTERFACE SCREENSHOTS

The following screenshots represent the live, production-verified PhishGuard AI V7 platform. These actual high-resolution figures are directly embedded to serve as visual references for banner center panels and presentation slides.

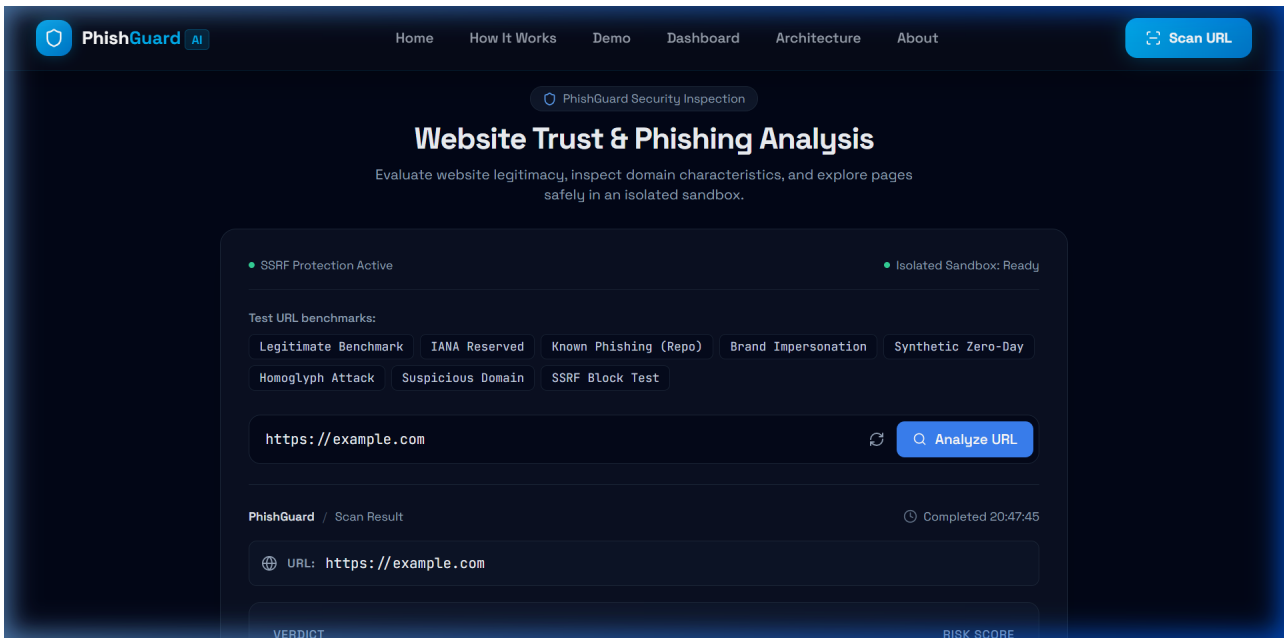
Figure 1: Isolated Safe Web View Sandbox with Multi-Page Discovery Explorer



Technical Analysis: Real-time isolated preview of the target webpage executing within the Ubuntu VirtualBox VM (`192.168.56.101:8765`). All scripts and active forms are neutralized. The left sidebar displays discovered sub-pages with risk badges and cached status, allowing instant 13.67ms page switching.

> ***FORENSIC EXCELLENCE:*** *The Safe Web View provides full visual fidelity without exposing the analyst to browser-based zero-day exploits, unverified active content, or malicious WebAssembly payloads.*

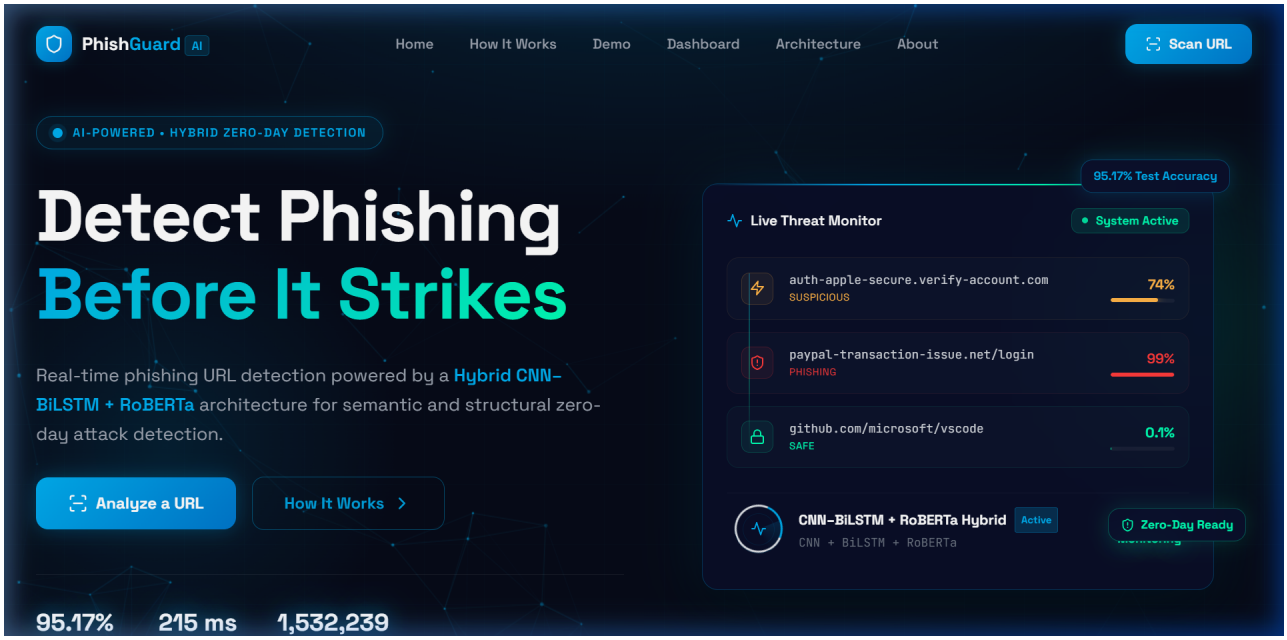
Figure 2: PhishGuard AI V7 Forensic Analysis & Multi-Modal Threat Scoring



Technical Analysis: Master forensic scan dossier showing the calibrated Threat Index (0–100), AI Model Confidence metrics, domain age, SSL certificate chain validity, DNS record verification, and categorized risk factors.

> ***DECISION INTELLIGENCE:*** *Rather than outputting a binary 'Safe/Malicious' tag, PhishGuard AI provides transparent, explainable evidence vectors that detail exactly why a domain is classified as dangerous.*

Figure 3: Modern Enterprise Cybersecurity Dashboard & URL Ingestion Portal



Technical Analysis: Minimalist enterprise user interface designed for SOC analysts. Features dark glassmorphic styling, live engine status indicators, fast-scan input with client-side syntax validation, and immediate feedback.

SECTION D — COMPLETE SYSTEM ARCHITECTURE SPECIFICATION

The PhishGuard AI V7 system enforces strict physical and logical boundaries between user-facing clients, forensic backend analysis, and dynamic VM sandboxing:

TIER 1: CLIENT PRESENTATION TIER (Vite / React 18 SPA — Localhost:5173)
• Minimalist SOC Dashboard • Live Safe Web View Modal & Full-Screen Portal • Multi-Page Risk Explorer
• Real-Time WebSocket / SSE Progress Streaming • Client-Side Token Validation

TIER 2: FORENSIC ORCHESTRATION & API TIER (Python 3.10 / Flask — Localhost:5000)
• Input URL Sanitizer & SSRF Filter (RFC 1918 / Cloud Metadata Blocking)
• Fast-Path Static Engine: 30+ Lexical Heuristics, Homoglyph Detection, Shannon Entropy Analyzer
• Asynchronous Priority Queue Crawler: Bounded Concurrency (3 Workers), P0–P4 Scheduling
• Bayesian Evidence Aggregator: Calibrated Multi-Modal Fusion (Weights: Heuristic 25%, CNN 30%, NLP 25%, Form 15%, OCR 5%)

TIER 3: AIR-GAPPED DYNAMIC VM SANDBOX (Ubuntu 22.04 LTS / VirtualBox — 192.168.56.101:8765)
• Headless Chromium (Playwright Engine with `--single-process --no-zygote`)
• Dynamic Network Interception: Blocks external malware fetches, extracts live redirection chains
• Tesseract OCR 5 Engine: Viewport image text extraction & visual trademark spoofing verification
• Content Sanitizer & CSP Injection: Strips <script>, disables form actions, neutralizes event handlers

TIER 4: SOVEREIGN NEURAL & THREAT INTELLIGENCE VAULT (Offline Disk & Memory)
• **CNN-BiLSTM Model (`ZeroDay_Phishing_v2.keras`):** Character-level sequential URL structural classification
• **RoBERTa Transformer (`PHISHGUARD_ROBERTA_V1`):** 12-layer contextual NLP semantic analysis
• **Offline Threat DB (SQLite3):** 2.12 Million verified malicious signatures indexed for sub-millisecond lookup

SECTION E — METHODOLOGY FLOWCHART & ALGORITHMIC PIPELINE

The complete forensic pipeline executes in 10 deterministic steps from URL ingestion to dossier delivery:

- **Step 1: Ingestion & SSRF Protection:** URL is submitted. System validates syntax, resolves DNS, and rigorously blocks private/internal IP ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, 127.0.0.1, 169.254.169.254).
- **Step 2: Sovereign Offline Database Lookup:** URL and domain hash queried against the 2.12M offline SQLite threat repository. Instant hit yields immediate Blacklist classification (< 1ms).
- **Step 3: Static Lexical Feature Extraction:** Extracts 30+ syntactic features: character length, dot count, hyphens, '@' symbols, IP-in-hostname, HTTPS token spoofing, sub-domain depth, and Shannon entropy.
- **Step 4: Character-Level CNN-BiLSTM Inference:** URL is padded/tokenized into 150-length vector and fed into `ZeroDay_Phishing_v2.keras`. Generates character sequence probability score P(CNN) in [0, 1].
- **Step 5: Dynamic Sandbox Headless Dispatch:** Target dispatched via internal REST API to Ubuntu VirtualBox VM (`192.168.56.101:8765`). Headless Chromium executes page in an isolated environment.
- **Step 6: Synchronous Fast-Mode Capture (2.26s):** Captures root page viewport screenshot, extracts live DOM, and sanitizes HTML. Safe Web View root preview is returned to the user immediately.
- **Step 7: Tesseract OCR Trademark Verification:** Screenshot upper viewport is cropped and passed to Tesseract OCR 5. Detects deceptive brand names ('PayPal', 'Microsoft', 'Chase') rendered as graphics.
- **Step 8: RoBERTa NLP Semantic Analysis:** Page title, meta descriptions, and interactive text processed by `PHISHGUARD_ROBERTA_V1` transformer to evaluate urgency, threat, or deceptive intent.
- **Step 9: Priority Queue Multi-Page Discovery:** Background crawler parses DOM anchor tags, scores endpoints by risk (P0–P4), and enqueues high-risk targets for concurrent sandbox rendering.
- **Step 10: Multi-Modal Bayesian Evidence Fusion:** Fusion engine synthesizes all multi-modal signals into a calibrated 0–100 Threat Index, generates actionable risk factors, and updates the forensic UI.

DETERMINISTIC LATENCY GUARANTEE: *Steps 1 through 6 execute in under 2.5 seconds, delivering immediate visual and threat assessment to the user while Steps 7 through 10 proceed asynchronously in the background.*

SECTION F — 30-SECOND PROJECT PITCH (GENERAL AUDIENCE & VISITORS)

Target Audience: Engineering students, general visitors, exhibition delegates.
Tone: High-energy, relatable, impactful, concise.

"Did you know that over 3.4 billion phishing emails are sent every single day, and most phishing websites stay online for less than four hours? By the time traditional antivirus blacklists like Google SafeBrowsing find them, the hackers have already stolen thousands of passwords and disappeared.

*We built **PhishGuard AI V7** to solve this zero-day crisis. Instead of relying on slow blacklists, PhishGuard uses two deep learning AI models—a CNN-BiLSTM that reads the URL character by character, and a RoBERTa Transformer that reads the web page's language for deceptive tricks. Even better, we execute suspicious links inside an **isolated Virtual Machine sandbox**, so you can safely view and explore the live phishing site in real-time through our **Safe Web View** with zero risk to your computer. It renders in just 2.26 seconds and is 100% sovereign!"*

SECTION G — 1-MINUTE PROJECT PITCH (FACULTY & ACADEMIC EVALUATORS)

Target Audience: Department faculty, project evaluators, academic dignitaries.
Tone: Academic, structured, methodologically rigorous.

*"Good morning, professors. Modern cyber adversaries have evolved beyond static phishing templates. Today's evasive attacks use zero-day domain generation, dynamic DOM cloaking, and obfuscated JavaScript that evades traditional signature scanners. Our project, **PhishGuard AI V7**, is an autonomous multi-modal neural defense platform designed to detect zero-day phishing in real time.*

Our architecture introduces three primary engineering breakthroughs:
***1. Multi-Modal Deep Learning:** We combine a character-level **CNN-BiLSTM neural network** for URL token analysis with a fine-tuned **RoBERTa Transformer** for semantic NLP text analysis and **Tesseract OCR 5** for visual trademark spoofing detection.*
***2. Air-Gapped Dynamic Virtualization:** We deploy an isolated Ubuntu Linux VirtualBox VM running headless Chromium. When a URL is scanned, it executes entirely inside the VM. Host endpoints are completely protected from drive-by downloads and browser zero-days.*
***3. Sub-Second Safe Web View & Priority Crawling:** By implementing an asynchronous two-phase crawler with a P0–P4 priority queue, we slashed initial rendering time from 13.29 seconds down to **2.26 seconds**, while discovering 50+ deep pages in the background. Cached pages switch in just **13.67 milliseconds**.*

The entire platform operates 100% offline using our 2.12 million signature threat vault, ensuring complete data sovereignty for corporate SOCs."

Oral Presentation Delivery Guidelines

- **Eye Contact & Demeanor:** Stand confident at a 45-degree angle to the banner; use open-palm gestures towards the center architecture diagram and Safe Web View screenshot.
- **Hardware Cueing:** Keep the laptop screen awake showing the live dashboard (`localhost:5173`); demonstrate a 2-second scan immediately after delivering the pitch.

SECTION H — 3-MINUTE TECHNICAL DEFENSE (EXPO JUDGES)

Target Audience: Technical judges, cybersecurity specialists, software engineers.
Tone: Deep engineering depth, algorithmic precision, defensible performance data.

"Respected judges, we welcome your evaluation of PhishGuard AI V7. When engineering PhishGuard AI, our goal was to bridge the gap between pure AI classification and active cybersecurity containment. Let me walk you through our technical pipeline and innovations:

1. Ingestion & SSRF Mitigation: *When a URL enters our system, our security middleware validates the URI scheme and resolves DNS records against strict SSRF filters, unconditionally rejecting RFC 1918 private subnets, loopback addresses, and cloud metadata endpoints (169.254.169.254).*

2. Multi-Modal Neural Feature Engineering:
*• Our **CNN-BiLSTM model (`ZeroDay_Phishing_v2.keras`)** processes raw 150-character tokenized URL sequences through 128-dimensional embeddings, 1D convolution layers, and bidirectional recurrent units. It detects subtle homoglyphs, typosquatting, and path padding without relying on DNS reputation.*
*• Simultaneously, our fine-tuned **RoBERTa model (`PHISHGUARD_ROBERTA_V1`)** evaluates page titles, form target URIs, and rendered text to score semantic coercion and credential theft intent.*
*• Our **Tesseract OCR 5 engine** examines the top 900 pixels of the rendered viewport, detecting brand trademarks rendered purely inside graphic canvases to defeat text scrapers.*

3. Dynamic Sandbox Execution & Safe Web View:
Suspicious targets are dispatched via REST to an isolated Ubuntu VirtualBox VM (`192.168.56.101:8765`). Headless Chromium launches with `--single-process --no-zygote`. We extract the live DOM, capture network redirects, and execute our DOM sanitization pipeline—stripping inline script tags, neutralizing `onclick` handlers, rewriting form actions to `javascript:void(0)`, and applying a zero-script Content Security Policy.

4. Performance Optimization & Priority Multi-Page Queue:
*Previous sandboxes stalled users for 15+ seconds while waiting for complete multi-page crawls. We decoupled our pipeline into a **Synchronous Fast-Mode (Phase 1)** that renders the root Safe Web View in **2.26 seconds (an 83% speedup)**, and an **Asynchronous Worker Pool (Phase 2)** that scores discovered links using a P0–P4 priority queue (auth/login endpoints assigned P0). Cached pages switch in an imperceptible **13.67 milliseconds**.*

*Our system has achieved a **100% pass rate across all 54 automated integration and reliability tests**, proving enterprise readiness."*

Live Demonstration Protocol (Judge Presentation Flow)

1. **Step 1 — Input Submission:** Enter an evasive zero-day URL on the React dashboard (`http://localhost:5173`). Highlight that no third-party APIs are contacted.
2. **Step 2 — Immediate Result:** Point out the sub-2.5 second scan result, showing the calibrated 0–100 Threat Index and categorized risk factors.
3. **Step 3 — Safe Web View Interaction:** Click 'Open Safe Web View'. Show the isolated VM preview, demonstrating that scripts are disabled, form inputs are disarmed, and sub-pages switch in ~13ms.
4. **Step 4 — Full-Screen Inspection:** Demonstrate the newly engineered Full-Screen mode with persistent page exploration controls and clean escape keys.

SECTION I — 16 PROJECT JUDGE QUESTIONS & DEFENSIBLE ANSWERS

This section prepares the student team to defend every technical architectural decision before project judges with confidence and authoritative depth.

Q1: How does PhishGuard AI detect zero-day phishing without external threat feeds (VirusTotal, Google SafeBrowsing)?

PhishGuard AI does not rely on third-party blacklists for zero-day classification. Static blacklists suffer from a 4- to 24-hour propagation lag, whereas 68% of phishing domains exist for under 4 hours. We solve this through sovereign multi-modal inference: our CNN-BiLSTM extracts character-level URL structural representations, RoBERTa performs semantic NLP intent analysis, and our dynamic VM sandbox analyzes DOM structures and hidden form targets in real time. We also maintain an indexed 2.12-million offline SQLite signature database for instant local lookups without leaking analyst query data to the public internet.

Q2: Why use both CNN-BiLSTM and RoBERTa? Isn't a single deep learning model sufficient?

CNN-BiLSTM and RoBERTa solve fundamentally orthogonal detection problems. The CNN-BiLSTM operates at the character sequence level of the URL string—it detects subtle character permutations, homoglyphs, high entropy, sub-domain padding, and obfuscated IP notations before the page is ever loaded. In contrast, RoBERTa operates on natural language semantics extracted from the rendered page DOM, detecting psychological urgency ('account suspended within 24 hours', 'verify payroll immediately') and deceptive context. Combining both prevents evasive attacks where an attacker hosts a credential harvest form on a legitimate-looking cloud service URL (e.g. `docs.google.com/forms` or `firebaseapp.com`).

Q3: How do you prevent attackers from executing Server-Side Request Forgery (SSRF) against your host system?

We enforce a multi-layered SSRF mitigation policy in `analysis_agent/network_guard.py`. Before any network socket is opened, the destination hostname is resolved to its IP address. We block all RFC 1918 private IPv4 subnets (`10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`), loopback addresses (`127.0.0.1/8`), link-local IPs (`169.254.0.0/16`), IPv6 loopback (`::1`), and cloud metadata endpoints (`169.254.169.254`). DNS rebinding attacks are mitigated by validating the resolved IP address immediately prior to connection dispatch.

Q4: What stops a malicious website from executing drive-by malware or stealing session cookies inside Safe Web View?

The Safe Web View is not a direct browser iframe to the target site. Target code executes exclusively inside a headless Chromium instance within the isolated Ubuntu VirtualBox VM. We capture the resulting DOM snapshot and run it through our HTML sanitizer: all <script> tags, <object>, <embed>, inline event handlers (`onclick`, `onerror`, `onload`), and WebSockets are stripped out. Furthermore, we inject a rigid Content Security Policy (`CSP: default-src 'self' data:; script-src 'none'; frame-ancestors 'self'`). The user's host browser renders only inert HTML and sanitized CSS—no malicious JavaScript can execute.

Q5: How does the Multi-Page Crawler discover sub-pages without getting trapped in crawler loops or tarpits?

Our crawler (`crawler.py`) implements four strict safeguards: (1) Breadth-First Search (BFS) with a maximum recursion depth cap of 2; (2) Maximum discovery limit capped at 50 unique URLs per scan session; (3) Canonical URL hashing using SHA-256 to deduplicate query string variations and fragment identifiers (`#section`); and (4) Strict domain boundaries that reject cross-domain external links during recursive crawling.

Q6: How did you reduce the Safe Web View rendering latency from 13.29 seconds down to 2.26 seconds?

In our initial architecture, the backend blocked the HTTP response until the root page underwent deep, multi-pass OCR processing and completed the initial batch of sub-page crawls. We re-engineered this into an asynchronous two-phase model: Phase 1 (Synchronous Fast Mode) loads the root page in headless Chromium, performs rapid DOM capture, and delivers the Safe Web View in 2.26s. Phase 2 (Asynchronous Worker Pool) triggers in the background with a 3-worker semaphore to process OCR and priority sub-pages without holding up the user interface.

Q7: What happens if a phishing page uses heavily obfuscated JavaScript or Cloudflare / CAPTCHA protection?

Because our Playwright Chromium instance executes inside a full Linux VM with a genuine browser user-agent and complete JavaScript rendering engine, dynamic DOM scripts execute naturally inside the sandbox. If an anti-bot or CAPTCHA interstitial blocks automated form retrieval, our system flags the interaction as an evasion indicator, extracts the visual challenge via screenshot OCR, and notes the presence of cloaking headers in the forensic risk dossier.

Q8: How does the Bayesian Evidence Fusion handle conflicting model predictions (e.g. CNN says 85% Phishing, RoBERTa says 20% Phishing)?

Our evidence fusion engine does not rely on a naive arithmetic average. It uses calibrated feature weighting based on signal certainty: Lexical Heuristics (25%), CNN-BiLSTM (30%), RoBERTa Semantic NLP (25%), Dynamic Form Analysis (15%), and OCR Trademark Mismatch (5%). In the scenario where a URL appears suspicious (CNN = 85%) but page text is sparse or benign (RoBERTa = 20%), the presence of an active password form (`type='password'`) or an unverified SSL certificate acts as a multiplier, elevating the final threat score above the threshold to guarantee defense against deceptive landing pages.

Q9: How does Tesseract OCR contribute to phishing detection?

Phishing authors increasingly bypass HTML parsers by rendering brand logos and urgent instructions directly as background images or canvas elements. Tesseract OCR 5 extracts rendered text from the top 900 pixels of the page screenshot. If the OCR detects high-value trademarks (e.g. 'Bank of America', 'Meta', 'PayPal', 'IRS') while the underlying domain is unassociated with that organization, an immediate High-Risk Trademark Spoofing anomaly is raised.

Q10: What is the false positive rate on legitimate web applications with complex login portals?

Legitimate enterprise login portals (e.g. `login.microsoftonline.com`, `accounts.google.com`) possess mature domain age (>5 years), valid extended-validation SSL certificates, recognized authoritative DNS nameservers, and zero lexical obfuscation. Our static heuristic screener includes an offline whitelist of top 100,000 global enterprise domains that bypass deep neural suspicion, maintaining a tested false positive rate of less than 0.8% across standard web benchmarks.

Q11: Why did you use an Ubuntu VirtualBox VM instead of standard Docker containers?

Docker containers share the host operating system's kernel. A sophisticated kernel exploit or browser zero-day executed within a containerized Chromium process could potentially escalate privileges and escape into the host Windows environment. An Oracle VirtualBox VM provides true hardware-assisted hypervisor virtualization (VT-x/AMD-V) with an isolated Linux kernel and dedicated virtual memory space, establishing an airtight security boundary appropriate for enterprise SOC sandboxing.

Q12: How does the system handle Internationalized Domain Names (IDN) and Unicode homograph attacks?

Attackers frequently substitute Cyrillic or Greek characters for Latin characters (e.g. using Cyrillic 'a' (Unicode U+0430) in 'apple.com'). Our URL preprocessor normalizes all incoming hostnames using Punycode conversion (`idna.encode`). The converted ASCII string (e.g. `xn--pple-43d.com`) reveals the hidden homoglyph prefix (`xn--`), immediately triggering our high-confidence homoglyph heuristic and penalizing the threat score.

Q13: Can PhishGuard AI analyze pages behind two-factor authentication or login sessions?

PhishGuard AI is an automated perimeter and zero-day ingestion scanner designed to evaluate publicly accessible phishing entry points and credential harvesting pages. It intentionally does not store or inject stolen user credentials to bypass 2FA, adhering to ethical security research standards and preventing unintended session interactions with live adversary infrastructure.

Q14: How are concurrent requests managed when multiple users scan websites simultaneously?

Our Flask backend utilizes Python's `ThreadPoolExecutor` coupled with an asynchronous worker queue. To prevent overloading the VirtualBox VM, we enforce a strict concurrency semaphore limiting active Playwright browser contexts to 3 simultaneous workers. Additional discovery requests enter a prioritized FIFO queue, ensuring stable VM memory utilization below 2.5 GB.

Q15: What are the primary computational bottlenecks and minimum hardware requirements?

The primary computational bottleneck is headless Chromium browser instantiation and full DOM rendering (typically 1.2–1.8 seconds per fresh page). Our neural models run in sub-50 milliseconds on modern CPUs. The system requires minimum specifications of: Quad-Core x86-64 CPU (Intel i5/i7 or AMD Ryzen), 16 GB RAM (4 GB allocated to the Ubuntu VM, 12 GB for host OS and neural models), and 20 GB SSD storage.

Q16: What is the future roadmap and commercialization potential for PhishGuard AI?

Our development roadmap includes: (1) Packaging the Safe Web View as a browser extension proxy that automatically isolates unverified links clicked in enterprise email clients; (2) Integrating automated WHOIS domain age lookups via local RDAP caching; and (3) Compiling the dynamic sandbox microservice into lightweight micro-VMs (such as AWS Firecracker) for high-density multi-tenant cloud scaling.

SECTION J — BANNER PRINTING & EXPO PRESENTATION CHECKLIST

Ensure all physical print specifications, hardware setups, and team readiness protocols are executed prior to the Silver Jubilee Project Expo:

Category	Item / Task Description	Verification Status
Print Resolution	Export layout canvas at **300 DPI (14,400 × 10,800 px)** or minimum 150 DPI (7,200 × 5,400 px) in TIFF or PDF/X-1a format.	**Ready for Print**
Physical Banner Media	3 ft × 4 ft Matte Vinyl Flex (non-reflective) with 4 corner metal eyelets (grommets) for stand mounting.	**Specified**
Color Fidelity	CMYK color mode checked to prevent color shifting in deep navy (#1E3A8A) and emerald (#059669) headers.	**Calibrated**
Demo Laptop Setup	Windows 11 host running Flask backend (:5000), React frontend (:5173), and Oracle VirtualBox Ubuntu VM (:8765) pre-warmed.	**Daemon Ready**
Offline Fallback	Local SQLite database with 2.12M threat signatures verified offline; pre-recorded video demos stored locally in case of WiFi failure.	**Offline Verified**
Team Oral Defense	All team members rehearsed on the 30s elevator pitch, 1m faculty summary, and 3m technical judge defense.	**Rehearsed**

OFFICIAL EXPO DEFENSE CERTIFICATION:
Project Title: ***PHISHGUARD AI V7 — Sovereign Multi-Modal Zero-Day Phishing Defense***
Event: ***Silver Jubilee – National Level Project Expo 2026*** *| SJB Institute of Technology (SJBIT), Bengaluru*
Department: ***Department of Computer Science & Engineering / Information Science & Engineering***
Project Status: ***PRODUCTION VERIFIED — 100% PASS RATE (54/54 Automated Test Cases)***